BRAC UNIVERSITY

Inspiring Excellence

MAT110

Differential Calculus and Coordinate Geometry

Assignment 02

Spring 24

Submission Date: 5th March, 2024 (Tuesday)

Instructions:
Assignment should be handwritten.Please write your Name and ID on the cover page of your assignment answer script.Make PDF of your assignment and submit it to the provided google form.

Solve all problems.Answer the questions by yourself. Plagiarism will lead to an F grade in the course. Each question carries 10 marks. Total marks is 60 and it will be converted to 15.

1. If $y = (cos^{-1}x)^2$ then show that,

$$\left(1 - x^2\right) y_{n+2} - (2n+1)\, x y_{n+1} - n^2 y_n = 0$$

By using Leibnitz's product rule.

2. Find the relative extrema (maxima/minima) of by using both first and 2nd derivative test.

$$f(x) = \frac{x^2}{x^4 + 16}$$

3. Use an appropriate linear approximation to find the value of $(1.97)^3$.Compare the approximation to the result produced directly by your calculating device.

4. Verify that hypotheses of Rolle's theorem are satisfied in the given interval. And find all values of c in that interval that satisfy the conclusion of the theorem.

$$f(x) = \cos x; \qquad [\frac{\pi}{2}, \frac{3\pi}{2}]$$

5. Find open intervals on which,

$$\frac{x}{x^2 + 2}$$

is (a) increasing, (b) decreasing, (c) concave up, (d) concave down. Also find all inflection points.

6. Find the Taylor Polynomials of orders $n = 0, 1, 2, 3, 4$ about $x = x_0$ and then find nth Taylor polynomial for the function in sigma notation.

$$f(x) = \frac{1}{x}, \qquad x_0 = -1$$